

Financial Econometrics

Professur Finance | Course introduction, WiSe 2026/27

Prof. Dr. Peter N. Posch
TU Dortmund

AGENDA

1. General information and plan of the semester
2. Mission
3. What is Financial Econometrics?
4. Examples: regression and time series
5. Statistical significance and tools

General information

Name	Financial Econometrics
Module	Master Finance
Credits / SWS	7.5 / 4
Lecturer	Prof. Dr. Peter N. Posch
TA	Alexander Krause
Start date	Tuesday, 13 October 2026



Lecture

Tuesdays, 14:15 – 15:45
Room M827



Tutorial

Tuesdays, 16:00 – 17:30
Room M827

What makes this lecture special?

What you will learn: how to answer economic questions using lots of data



Econometrics



Python in Jupyter Notebooks



GitHub



Teamwork



Mission for an exam

Plan of the semester

Week	Date	Topic	Lecturer	Tutorial
1	13.10.2026	Welcome, Regression MLE	PNP	PNP
2	20.10.2026	Python “old school” + Coding Agents	AK	AK – Coding Agents with Claude
3	27.10.2026	Python “old school” + Coding Agents		
4	03.11.2026	Time Series 1	PNP	AK
5	10.11.2026	Time Series 2	PNP	AK
6	17.11.2026	Simulation / MC / Bootstrap	PNP	AK
7	24.11.2026	Energy Finance	AK	AK
8	01.12.2026	Energy Markets & Trading	PNP	AK
9	08.12.2026	Presentation Dr. Miguel Krause (EON) + Presentation of the missions (Registration for exam 01.12. – 16.12.)	PNP	AK
10	15.12.2026	Registration Missions + Q&A regarding Missions	PNP / AK	AK – Missions
22.12. and 29.12.2026 – winter break, no sessions				
11	05.01.2027	Q&A	AK	–
12	12.01.2027	Q&A	AK	–
13	19.01.2027	Presentation Missions (Deadline of mission submission 3 days before)		
14	26.01.2027	Backup presentation date		
15	02.02.2027	State Space Models + Kalman Filter	PNP	

PNP = Prof. Dr. Peter N. Posch, AK = Alexander Krause

Mission – your exam



1 Notebook + Dataset

Jupyter Notebook in Python, no written paper. Explain your code and interpret your results in Markdown text.



2 Presentation

Five PowerPoint slides and an oral presentation of 15 minutes. You can find a PowerPoint template in the Moodle course.

The exam consists of two elements – the final presentation takes the place of an oral exam.

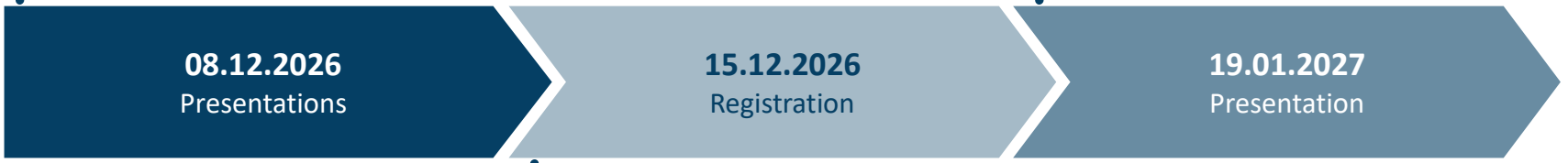
Mission – details

1. **Choose a paper** whose results (tables & figures) can be reproduced in Python.
2. **Get inspired:** in December, examples of possible papers will be presented. You can choose one of them.
3. **Your own paper:** possible if approved in advance by us – but we strongly recommend one of the papers presented in December, as no support can be provided if your own reproduction fails.
4. **Teamwork:** max. 2 group members. You may also work alone, but we recommend working in a group.
5. **Registration for missions:** 15.12.2026.

Mission timeline

Presentation Miguel + Presentation of the missions.

Present your mission: 5 slides, 15 minutes. Backup date: 26.01.2027.



Registration for missions (max. 2 group members). Q&A sessions on 05.01. and 12.01.2027.

Mission example: your notebook

Not only code – explain your work

- **Explanations:** describe the content and each step as Markdown text between the code cells
- **Structure:** title, table of contents and one section per step
- **Interpretation:** discuss your results as Markdown text inside the notebook
- **Example:** reproduction of the three-factor regression of Fama & French (1993)

Reproducing Fama & French (1993)

Three-factor model for the small / low book-to-market portfolio, 1963-07 to 1991-12

Contents: 1. Data · 2. Regression · 3. Interpretation

1. Data

Monthly returns from the Kenneth French data library. The dependent variable is the **excess return** of the portfolio (portfolio return minus risk-free rate).

In [1]:
```
import statsmodels.formula.api as smf
from linearmodels.datasets import french

df = french.load().set_index("dates").loc["1963-07":"1991-12"]
df["excess"] = df["S1V1"] - df["RF"]
```

2. Regression

We regress the excess return on the market (MktRF), size (SMB) and value (HML) factors.

In [2]:
```
res = smf.ols("excess ~ MktRF + SMB + HML", data=df).fit()
print(f"R-squared = {res.rsquared:.3f}")
res.summary2().tables[1].iloc[:, :4].round(3)
```

```
R-squared = 0.939
```

Out[2]:

	Coef.	Std.Err.	t	P>|t|
Intercept	-0.004	0.001	-3.627	0.0
MktRF	1.036	0.026	39.408	0.0
SMB	1.408	0.039	35.949	0.0
HML	-0.288	0.044	-6.532	0.0

3. Interpretation

- **Market beta ≈ 1.04:** the portfolio moves roughly one-for-one with the market.
- **SMB = 1.41 (t = 35.9):** strong small-firm exposure, as expected for a small-cap portfolio.
- **HML = −0.29 (t = −6.5):** negative value loading, i.e. a growth tilt (low book-to-market).
- **R^2 = 0.94:** the three factors explain most of the variation in the portfolio's returns.

Example notebook: own run with data from the Kenneth R. French Data Library (via the Python package linearmodels).

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Introduction to Financial Econometrics

What is Financial Econometrics?

Definitions

Econometrics
= Application of statistical methods to economic data

Financial Econometrics
= Application of statistical methods to financial market data

Finance domains involved:
- Risk Management
- Quantitative Trading
- Portfolio Management

Example questions

- Do future markets react faster to news than spot markets?
- How to forecast the stock returns of different countries?
- Is CAPM a good model for returns of risky assets?
- How can credit ratings be determined?
- How to model the volatility of returns?
- How can we identify volatility clusters?
- Are financial markets informationally efficient?

What makes financial data special?



Frequency

Prices on financial markets change very frequently, e.g. per second. In economics they are typically measured at much lower frequency, e.g. quarterly GDP change.



Quantity

Correlated with frequency, the quantity of data produced is immense.

Amount of data created is increasing rapidly.



Quality

The recorded price of a transaction (trade) is the price at which the underlying changed hands.

Data, models and parameter estimation

Data

- Cross-sectional data
- Panel data
- Time series data

Models

Regression models

- linear
- nonlinear
- univariate
- multivariate

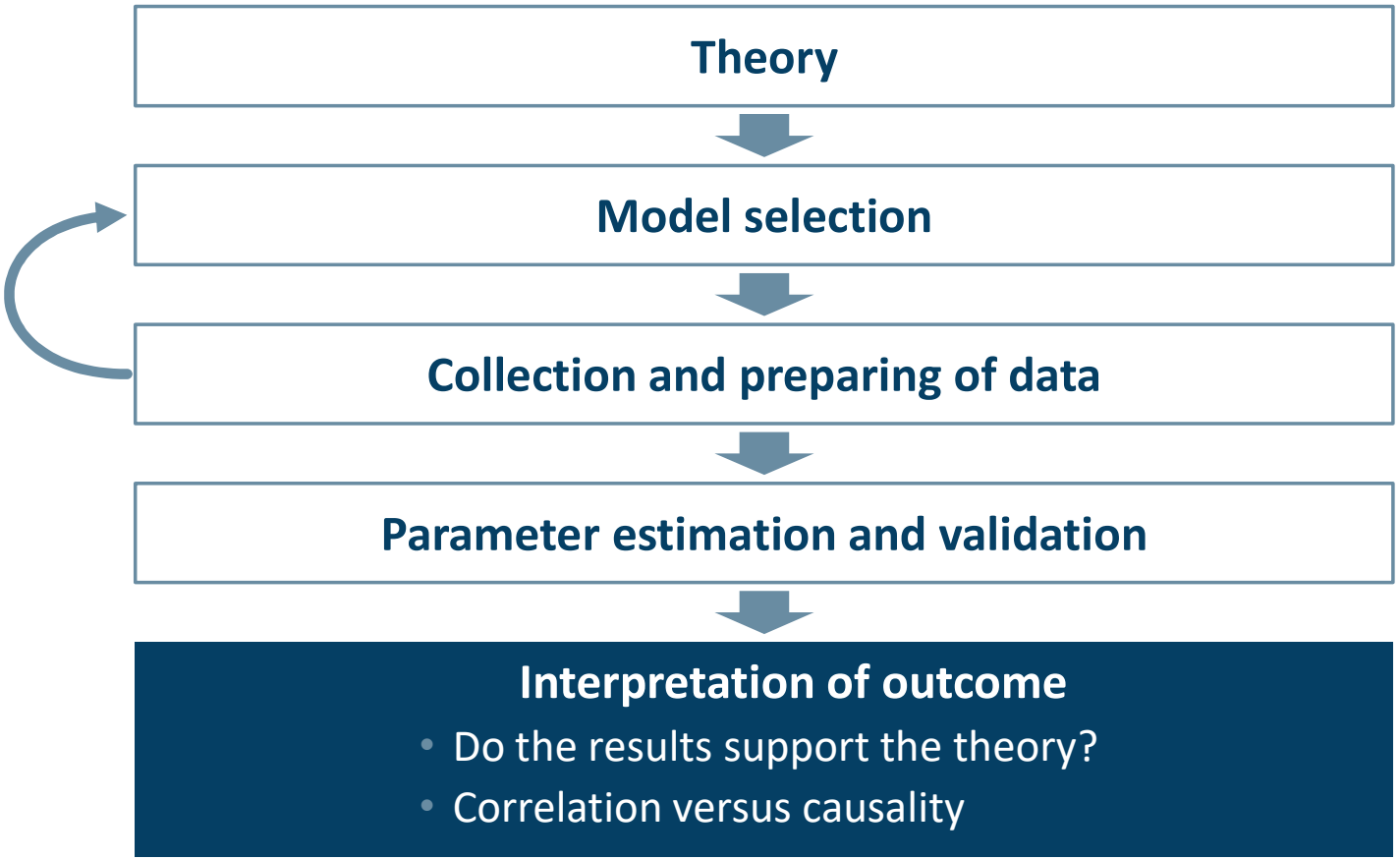
Time series models

- ARCH, GARCH
- ARIMA
- VAR

Estimation

- Maximum Likelihood Estimation (MLE)
- Least Squares (LS)
- Bayesian Methods (e.g. MCMC)

Formulation of an econometric model



Example I

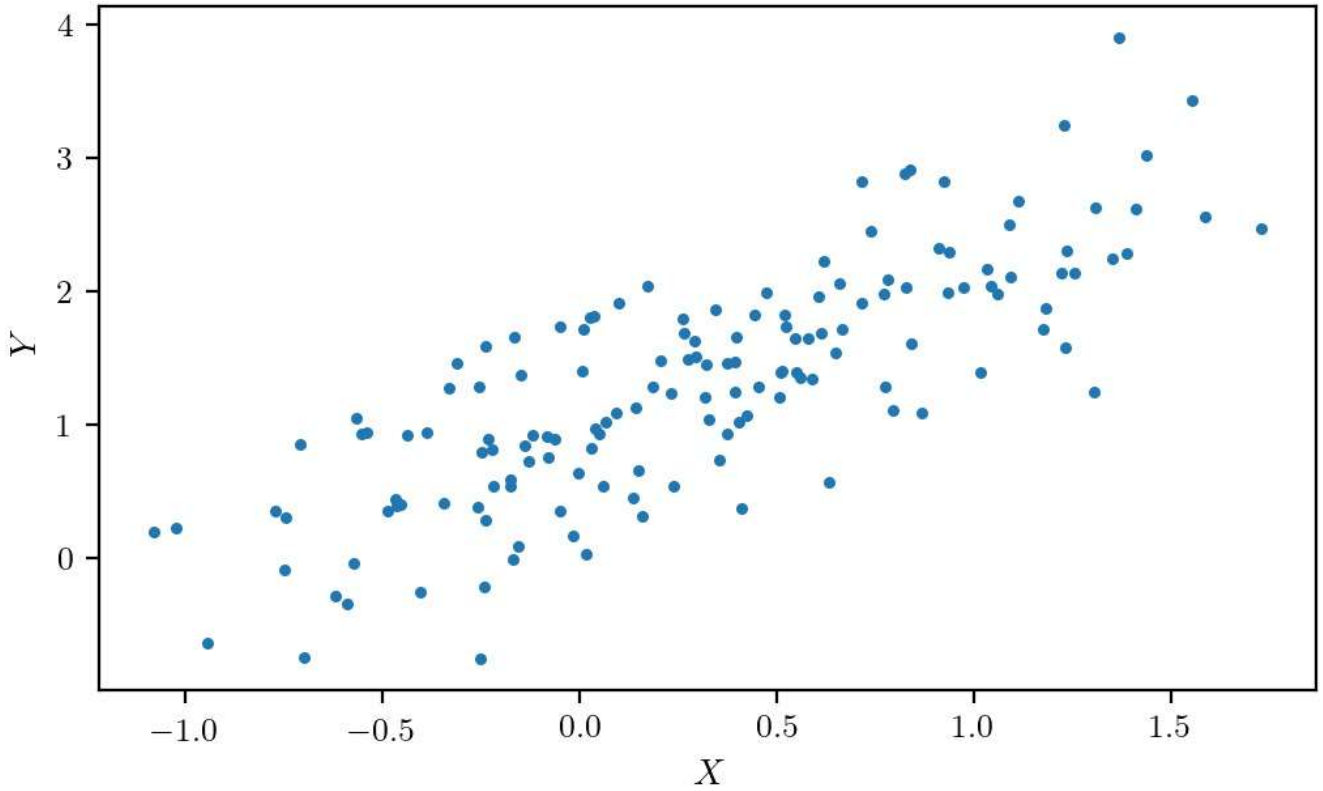
#	X	Y
1	-0.48	-0.09
2	-0.72	1.72
3	0.31	0.78
4	1.83	1.64
...		
...		
4999	0.08	0.76
5000	0.97	1.33

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Data and model selection

- Cross-sectional data
- *Assume* linear relationship between variables
- Choose linear regression model

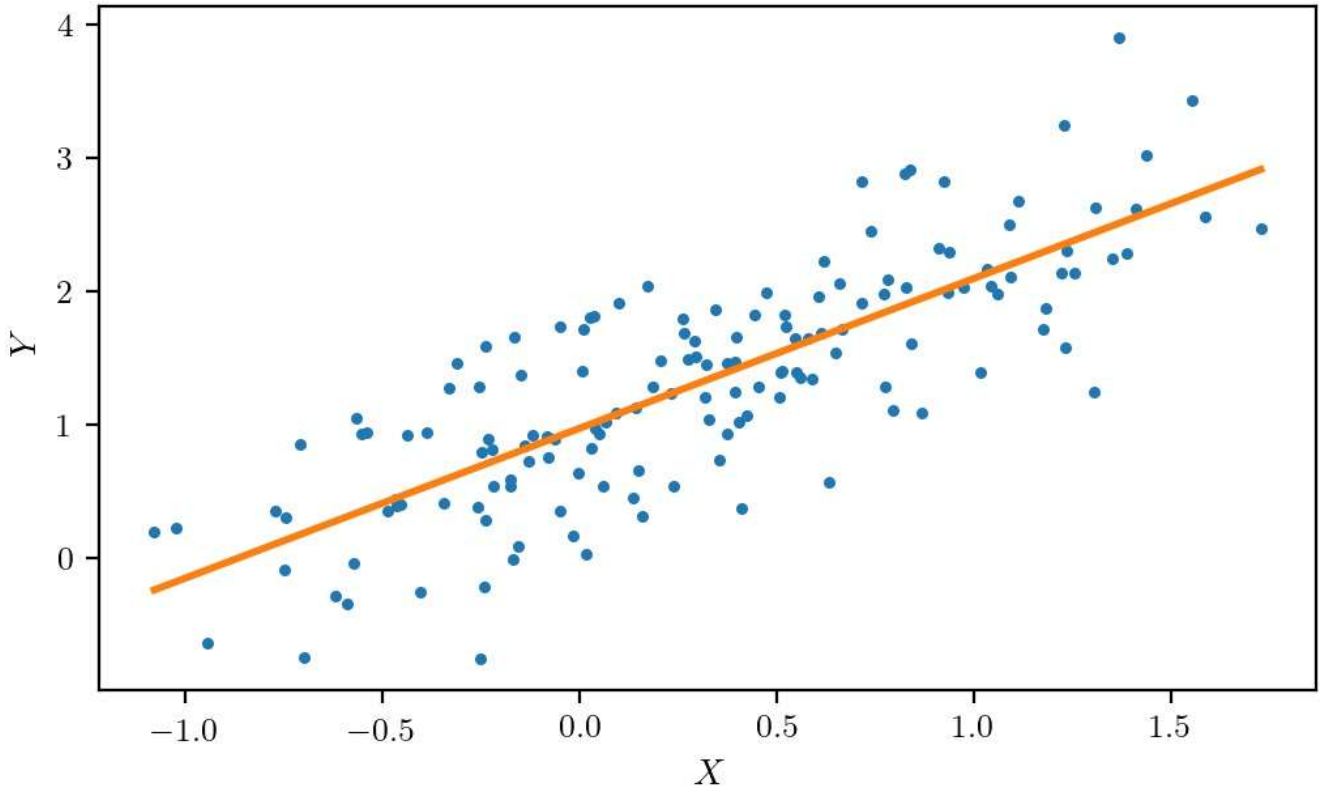


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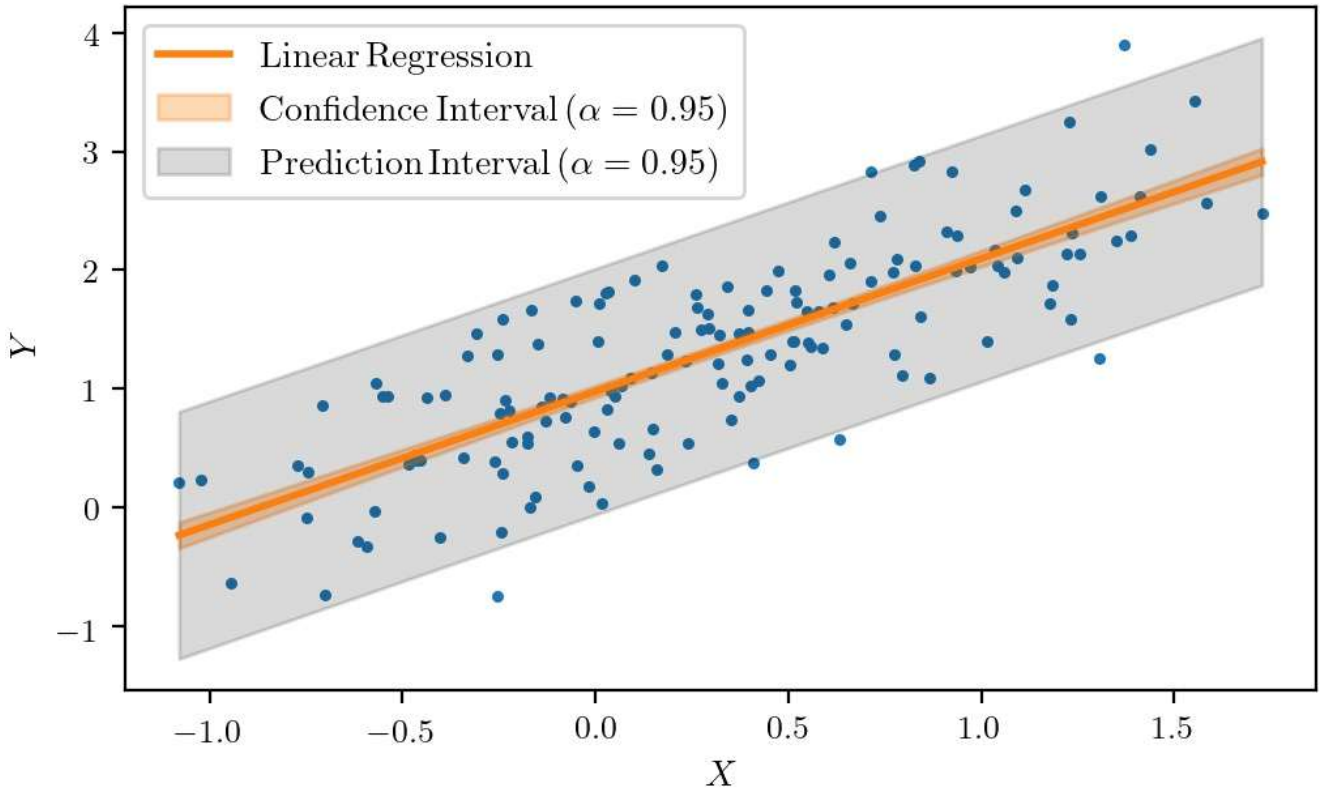


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Data and model selection

- Cross-sectional data
- *Assume* linear relationship between variables
- Choose linear regression model
- Every fit or prediction is associated with **uncertainties**



Example II

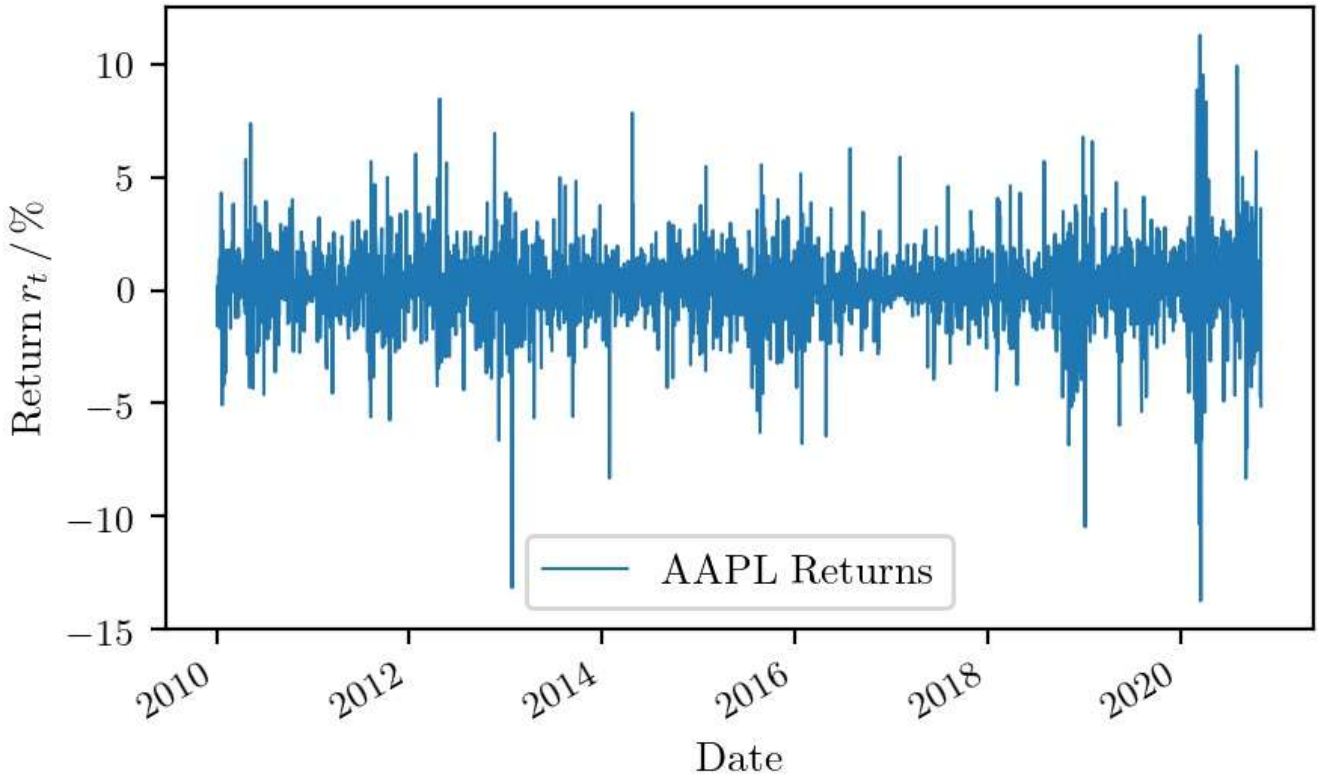
t	Return / %
2010-01-05	-0.16
2010-01-06	-1.60
...	
...	
2020-10-29	3.64
2020-10-30	-5.12

Example II

t	Return / %
2010-01-05	-0.16
2010-01-06	-1.60
...	
...	
2020-10-29	3.64
2020-10-30	-5.12

Data and model selection

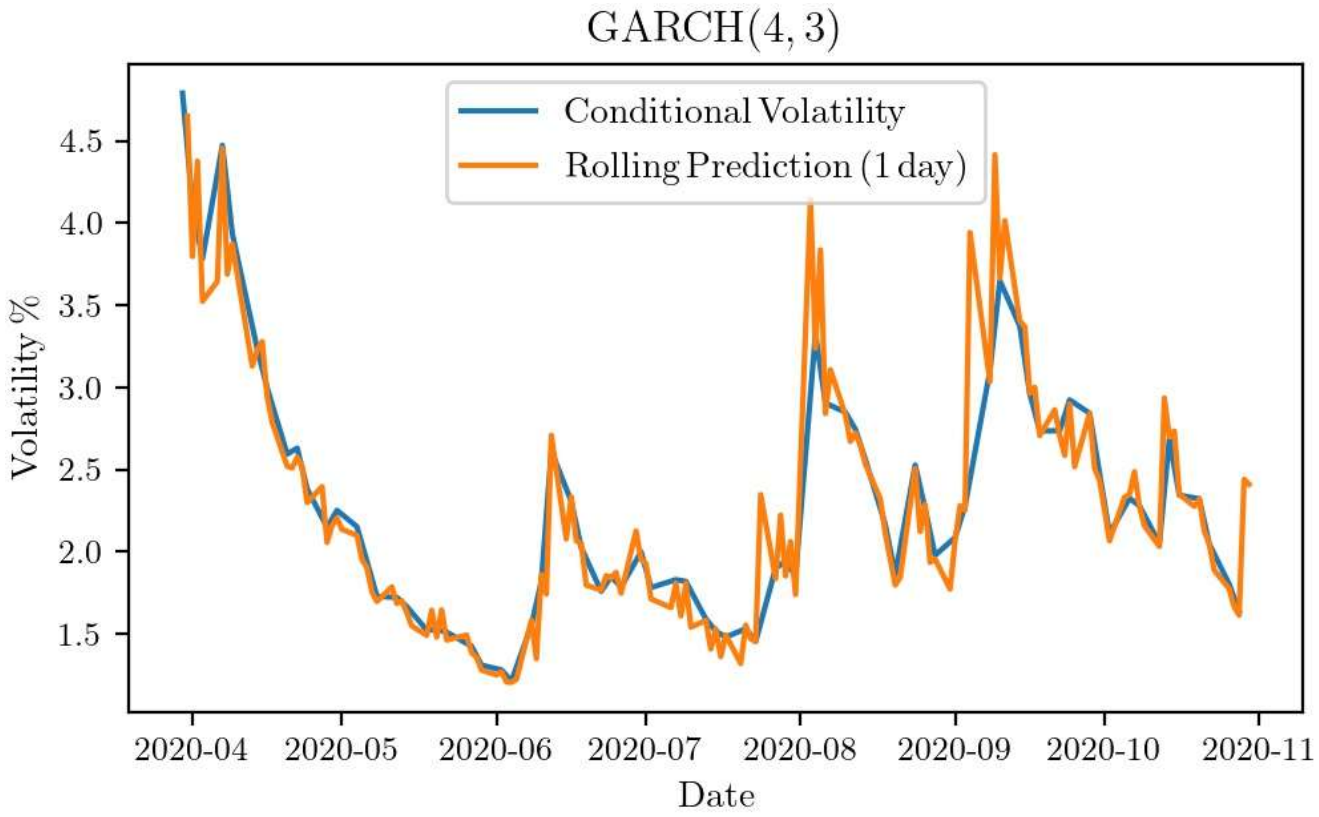
- Time Series data
- *Observe* volatility cluster
- Assume heteroskedasticity of returns
- Fit GARCH model to data to estimate conditional volatility



Example II

GARCH models

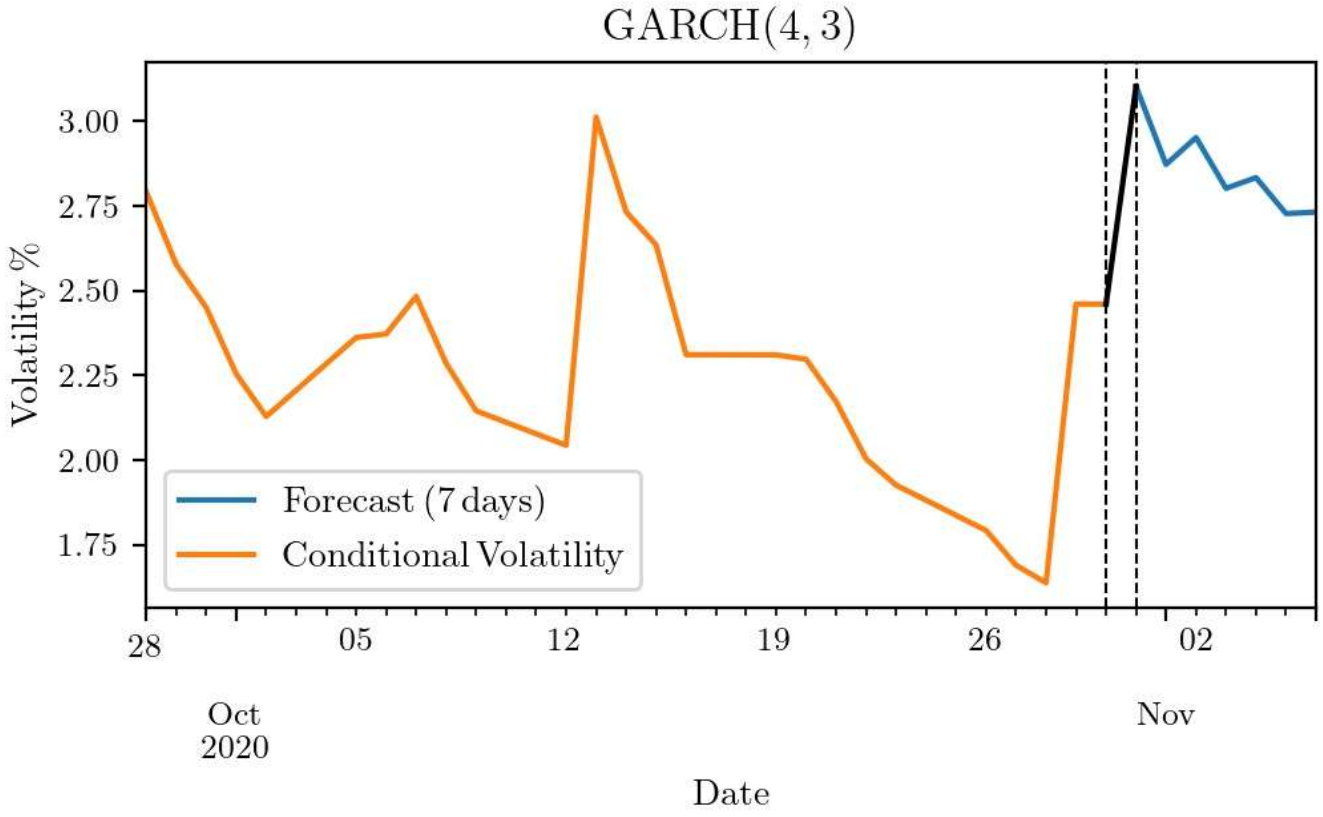
- Used to estimate volatility of returns for stocks or bonds
- Can be used to predict/forecast volatility



Example II

GARCH models

- Used to estimate volatility of returns for stocks or bonds
- Can be used to predict/forecast volatility
- Useful in portfolio risk analysis

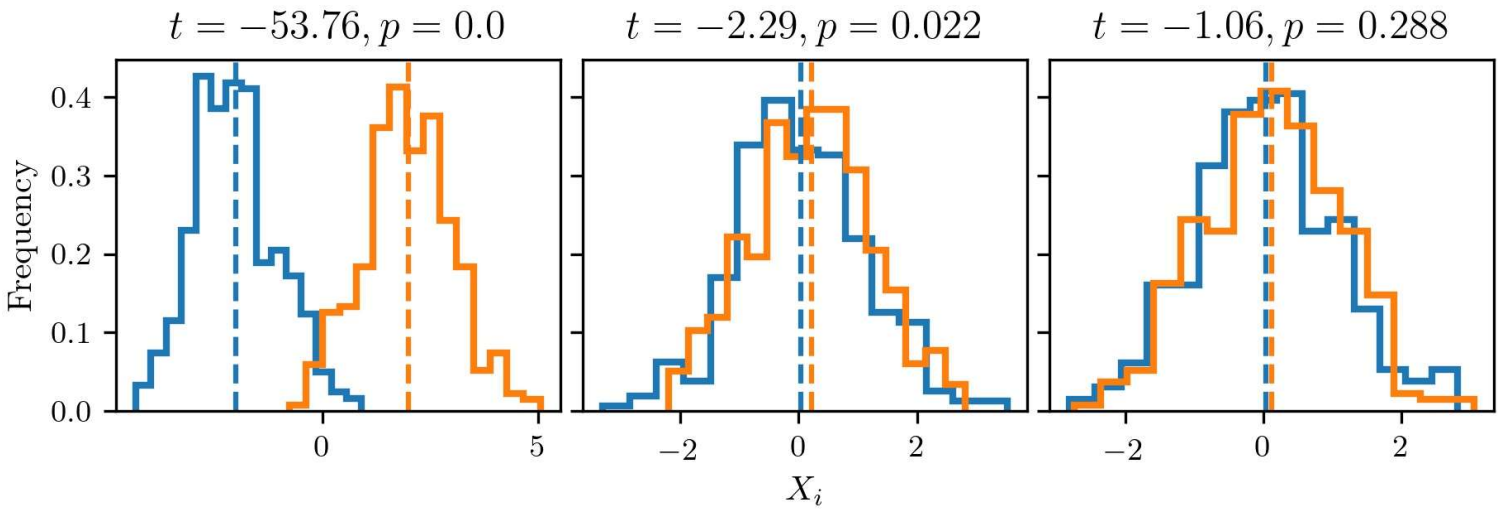


Statistical significance

- Quantify whether a result is likely due to chance or to some factor of interest, i.e. not likely to occur randomly

Example: t-test

- Null hypothesis states that both means of the normally distributed populations (orange and blue) are equal
- The test statistic corresponds to a p-value under given null hypothesis



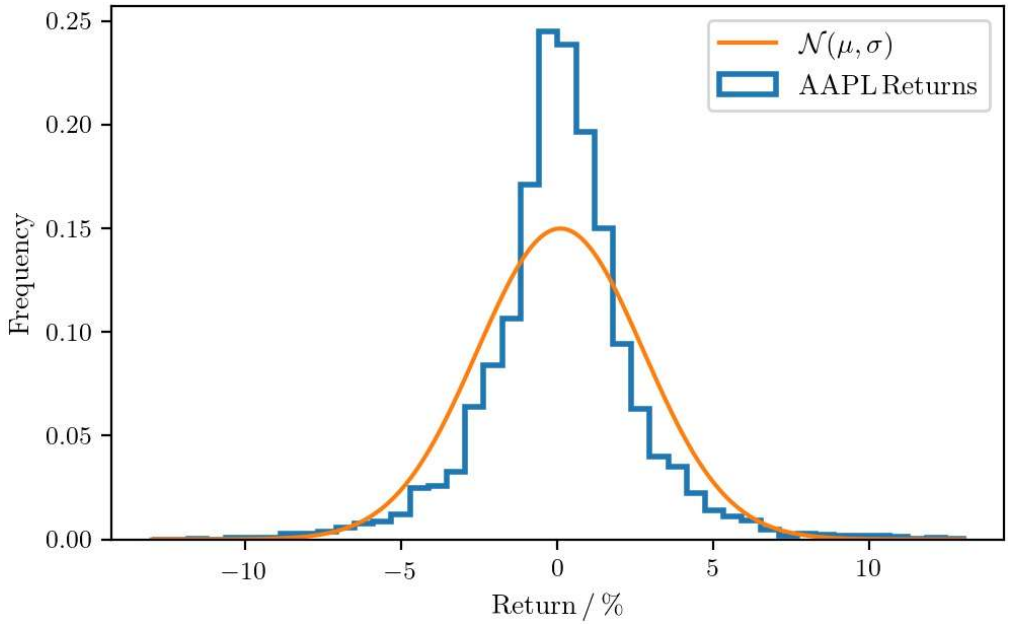
“All models are wrong, but some are useful”

Statistical models and tests are based on assumptions

- These assumptions hold true only in the fewest cases
- This results in a biased model

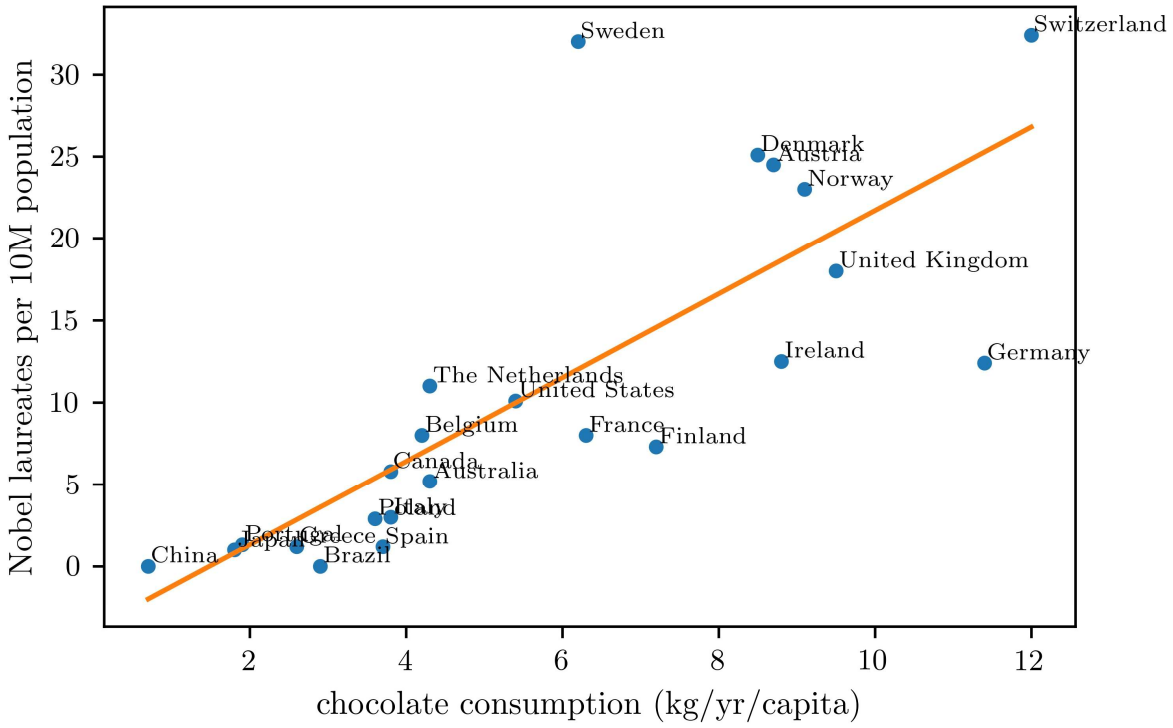
Examples:

- Linearity
- Stationarity
- Normality
- Omitted variables
- Survivorship bias



Correlation versus causation

- Correlation measures the relationship between two variables
- **But:** correlation does not imply causation



Source: Messerli, F. H. (2012). Chocolate consumption, cognitive function, and Nobel laureates. NEJM, 367(16), 1562–1564 [1]

Tools: requirements and challenges

Challenges

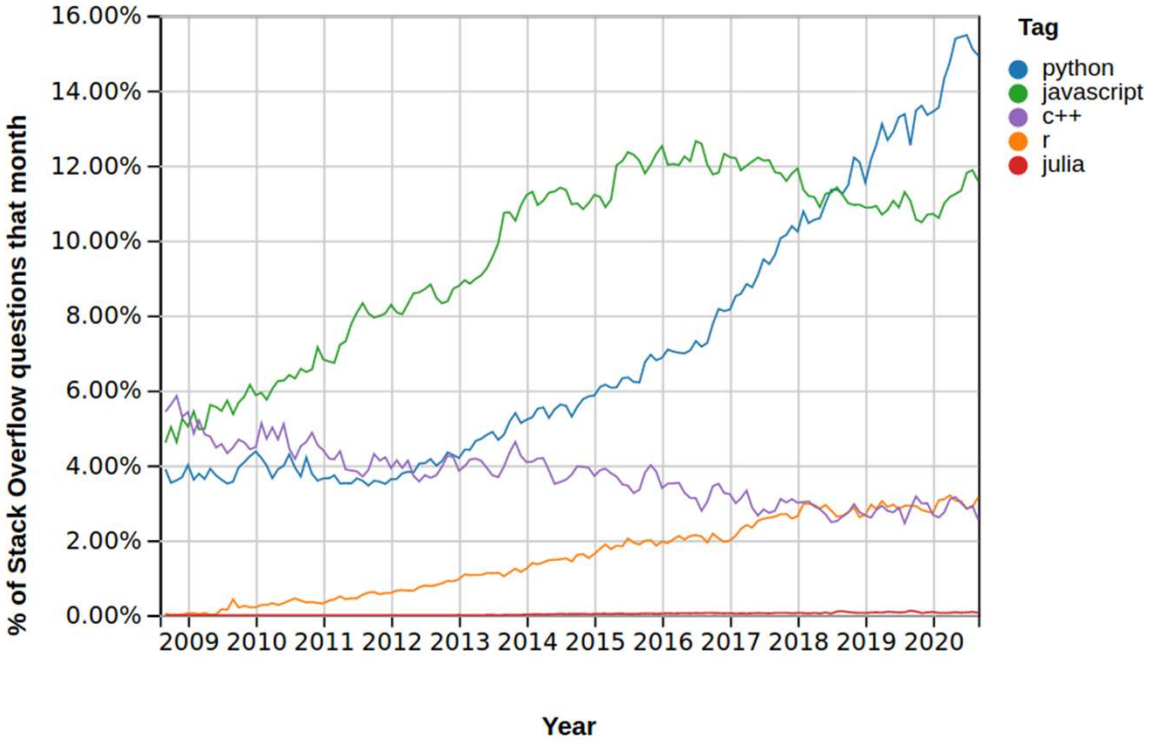
- Amount of data can be **huge**
- Integration with other software necessary
  - Databases

Requirements

- Efficient and scalable tools
- Fast computation
  - Parallel computation
  - GPU accelerated computation
- Easy to reproduce and automation of tasks
- Open source: inspect source code and modify software to your needs

Tools: why Python?

- Python is one of the most popular programming languages
- Many open-source data science libraries



Source: Stack Overflow Trends, https://insights.stackoverflow.com/trends [2]

References



[1] Messerli, F. H. (2012). Chocolate consumption, cognitive function, and Nobel laureates. The New England Journal of Medicine, 367(16), 1562–1564.

[2] Stack Overflow Trends. https://insights.stackoverflow.com/trends

[3] Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. Journal of Financial Economics, 33(1), 3–56.

[4] Kenneth R. French Data Library (monthly factor and portfolio returns), as distributed with the Python package linearmodels.